B3 — Quantum, Atomic and Molecular Physics — Model Solutions, 2017

Question 1 — Spin-orbit interaction in hydrogenic atoms

Problem. Derive the spin-orbit splitting for a hydrogenic atom and obtain $\Delta E = Z^4\alpha^2 hcR_\infty/[n^3 l(l+1)]$. Apply numerically to $n = 2, 4$ of H, draw the level diagram for the Balmer-β transition, and assess feasibility.

Derivation of ΔE ([9])

Electron in rest frame sees motional magnetic field from nuclear Coulomb field. Boost (non-relativistic): $\mathbf{B}' = -\mathbf{v}\times\mathbf{E}/c^2$.

Central field $\mathbf{E} = -(1/e)\,\nabla V = (\mathbf{r}/r)(1/e)\,\mathrm{d}V/\mathrm{d}r$ with $V(r) = -Ze^2/(4\pi\epsilon_0 r)$:

$$\mathbf{B}' = -\frac{1}{c^2 e}\,\frac{1}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\,(\mathbf{v}\times\mathbf{r}).$$

Substitute $\mathbf{L} = m_e\,\mathbf{r}\times\mathbf{v}$, so $\mathbf{v}\times\mathbf{r} = -\mathbf{L}/m_e$:

$$\mathbf{B}' = \frac{1}{m_e c^2 e}\,\frac{1}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\,\mathbf{L}.$$

Electron magnetic moment: $\mu_s = -g_s\,\mu_B\,\mathbf{S}/\hbar$. Interaction $H = -\mu_s\cdot\mathbf{B}'$. Apply Thomas correction $g_s \to g_s - 1 \approx 1$:

$$H_{\mathrm{SO}} = \frac{1}{m_e c^2 e}\,\frac{1}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\,\frac{\mu_B}{\hbar}\,\mathbf{L}\cdot\mathbf{S}.$$

Insert $\mu_B = e\hbar/(2m_e)$ and $(1/r)\,\mathrm{d}V/\mathrm{d}r = Ze^2/(4\pi\epsilon_0)\cdot 1/r^3$:

$$H_{\mathrm{SO}} = \frac{1}{2m_e^2c^2}\,\frac{Ze^2}{4\pi\epsilon_0}\,\frac{1}{r^3}\,\mathbf{L}\cdot\mathbf{S}. \tag{1}$$

First-order shift: $\Delta E_j = \xi_{nl}\,\langle\mathbf{L}\cdot\mathbf{S}\rangle$ with $\xi_{nl} = (\hbar^2/2m_e^2c^2)(Ze^2/4\pi\epsilon_0)\langle 1/r^3\rangle$.
Diagonal in j: $\langle\mathbf{L}\cdot\mathbf{S}\rangle = \frac{\hbar^2}{2}[j(j+1) - l(l+1) - s(s+1)]$.
Two values $j = l \pm \frac{1}{2}$ for $l \neq 0$. Difference:

$$\Delta\langle\mathbf{L}\cdot\mathbf{S}\rangle = \tfrac{\hbar^2}{2}\big[(l+\tfrac{1}{2})(l+\tfrac{3}{2}) - (l-\tfrac{1}{2})(l+\tfrac{1}{2})\big] = \hbar^2\big(l+\tfrac{1}{2}\big).$$

Splitting:

$$\Delta E = \xi_{nl}\,\hbar^2\big(l+\tfrac{1}{2}\big).$$

Substitute given $\langle 1/r^3\rangle = 2Z^3/[n^3 a_0^3 l(l+1)(2l+1)]$:

$$\Delta E = \frac{\hbar^2}{2m_e^2c^2}\,\frac{Ze^2}{4\pi\epsilon_0}\,\frac{2Z^3}{n^3a_0^3 l(l+1)(2l+1)}\,\hbar^2(l+\tfrac{1}{2}).$$

Use $2(l+\frac{1}{2}) = (2l+1)$ and cancel:

$$\Delta E = \frac{\hbar^4}{m_e^2 c^2}\,\frac{Z^4 e^2}{4\pi\epsilon_0 a_0^3}\,\frac{1}{n^3 l(l+1)}.$$

Substitute $a_0 = 4\pi\epsilon_0\hbar^2/(m_e e^2)$ once:

$$\Delta E = \frac{Z^4 e^4}{(4\pi\epsilon_0)^2}\,\frac{1}{m_e c^2 \hbar^2}\,\frac{1}{n^3 l(l+1)}\cdot\frac{1}{a_0}.$$

Rewrite using $\alpha = e^2/(4\pi\epsilon_0\hbar c)$ and $hcR_\infty = m_e c^2\alpha^2/2$:

$$\boxed{\Delta E = \frac{Z^4\alpha^2 hcR_\infty}{n^3 l(l+1)}} \tag{2}$$

Numerical evaluation for H ([5])

Constants: $\alpha^2 = (1/137.036)^2 = 5.325\times10^{-5}$, $cR_\infty = 3.290\times10^{15}\,\text{Hz}$.

Prefactor for H ($Z=1$):

$$\alpha^2 cR_\infty = 5.325\times10^{-5}\times3.290\times10^{15}\,\text{Hz} = 1.752\times10^{11}\,\text{Hz} = 175.2\,\text{GHz}.$$

2p ($n=2, l=1$):

$$\Delta\nu(2p) = \frac{175.2}{2^3\cdot1\cdot2}\,\text{GHz} = \frac{175.2}{16}\,\text{GHz} = \boxed{10.95\,\text{GHz}}.$$

Magnetic field on electron from $\Delta E = g_s\mu_B B\cdot\frac{1}{2}\cdot 2 = 2\mu_B B$ (separation of $j = l\pm\frac{1}{2}$ in field B is $2\mu_B B$ for $g_s = 2$, $\Delta m_s = 1$, but here SO splits via $\mathbf{L}\cdot\mathbf{S}$, equivalent B defined from $\Delta E = g_s\mu_B B$):

$$B = \frac{\Delta E}{g_s\mu_B} = \frac{h\,\Delta\nu}{2\mu_B}.$$

Numerical:

$$B = \frac{6.626\times10^{-34}\times1.095\times10^{10}}{2\times9.274\times10^{-24}}\,\text{T}.$$

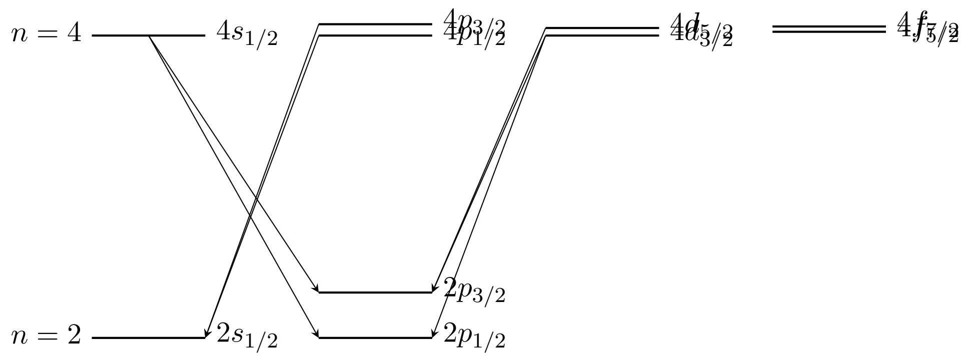
$$B = \frac{7.255\times10^{-24}}{1.855\times10^{-23}}\,\text{T} = \boxed{0.39\,\text{T}}.$$

Other configurations ($\Delta\nu = 175.2/[n^3 l(l+1)]$ GHz):

config	$n^3 l(l+1)$	$\Delta\nu$ / GHz
2s ($l=0$)	—	no SO splitting
2p ($l=1$)	16	10.95
4s ($l=0$)	—	no SO splitting
4p ($l=1$)	128	1.37
4d ($l=2$)	384	0.456
4f ($l=3$)	768	0.228

Energy-level diagram for Balmer-β ([6])

Levels labelled nl_j. Selection rules: $\Delta l = \pm 1$, $\Delta j = 0, \pm 1$ (no $j = 0 \to 0$).



Note: $4f \to 2$ is forbidden ($\Delta l = 3$). Allowed transitions: $4s \to 2p_{1/2,3/2}$; $4p_{1/2,3/2} \to 2s_{1/2}$; $4d_{3/2} \to 2p_{1/2,3/2}$; $4d_{5/2} \to 2p_{3/2}$.

Wavelength, Doppler width, feasibility ([5])

Balmer-β ($n = 4 \to 2$):

$$\frac{1}{\lambda} = R_\infty\Big(\frac{1}{4} - \frac{1}{16}\Big) = R_\infty \cdot \frac{3}{16}.$$

$$\lambda = \frac{16}{3R_\infty} = \frac{16}{3 \times 1.097 \times 10^7}\,\mathrm{m} = \boxed{486.1\,\mathrm{nm}}.$$

Doppler FWHM (H discharge $T \sim 300\,\mathrm{K}$):

$$\frac{\Delta\nu_D}{\nu_0} = \sqrt{\frac{8k_BT\ln 2}{mc^2}}.$$

Compute $k_BT = 4.14 \times 10^{-21}\,\mathrm{J}$, $m_Hc^2 = 1.503 \times 10^{-10}\,\mathrm{J}$:

$$\frac{8k_BT\ln 2}{mc^2} = \frac{8 \times 4.14 \times 10^{-21} \times 0.693}{1.503 \times 10^{-10}} = 1.527 \times 10^{-10}.$$

$$\frac{\Delta\nu_D}{\nu_0} = 1.236 \times 10^{-5}.$$

Frequency $\nu_0 = c/\lambda = 6.17 \times 10^{14}\,\mathrm{Hz}$:

$$\Delta\nu_D = 1.236 \times 10^{-5} \times 6.17 \times 10^{14}\,\mathrm{Hz} = \boxed{7.6\,\mathrm{GHz}}.$$

Comparison to fine-structure splittings: 2p splitting 11 GHz is comparable to Doppler width; 4-shell splittings 0.2–1.4 GHz are buried beneath it. Direct optical observation requires Doppler-free spectroscopy (saturated absorption or two-photon).

Question 2 — LS-coupling, Zeeman effect in mercury

Problem. Define LS-coupling, list silicon $3p\,4p$ terms, derive g_J, state E1 selection rules, and analyse Zeeman patterns of three Hg lines.

LS-coupling and Si $3p\,4p$ terms ([7])

LS-coupling: residual Coulomb (electron-electron) interaction $\gg$ spin-orbit. Couple individual orbital angular momenta to total $\mathbf{L} = \sum \mathbf{l}_i$ and spins to $\mathbf{S} = \sum \mathbf{s}_i$, then $\mathbf{J} = \mathbf{L} + \mathbf{S}$. Valid for light atoms (Z small) where $\xi_{nl}\langle \mathbf{l}\cdot\mathbf{s}\rangle \ll$ Coulomb term splittings.

Si: closed shells $1s^2 2s^2 2p^6 3s^2$ contribute $L = S = 0$. Open shells $3p\,4p$: $l_1 = l_2 = 1$, $s_1 = s_2 = \frac{1}{2}$. Different n, so no Pauli restriction.

$L = 0, 1, 2$; $S = 0, 1$. Terms:

$$\boxed{{}^1S_0,\ {}^1P_1,\ {}^1D_2,\ {}^3S_1,\ {}^3P_{0,1,2},\ {}^3D_{1,2,3}.}$$

Landé g_J ([6])

Magnetic moment: $\mu = -\mu_B(\mathbf{L} + g_s\mathbf{S})/\hbar$ with $g_s \simeq 2$:

$$\mu = -\frac{\mu_B}{\hbar}(\mathbf{L} + 2\mathbf{S}) = -\frac{\mu_B}{\hbar}(\mathbf{J} + \mathbf{S}).$$

Project on $\mathbf{J}$ (vector model, $\mathbf{L}, \mathbf{S}$ precess about $\mathbf{J}$):

$$\langle \mu \rangle = -\frac{\mu_B}{\hbar}\Big(1 + \frac{\langle \mathbf{J}\cdot\mathbf{S}\rangle}{\langle J^2\rangle}\Big)\,\mathbf{J}.$$

Compute $\mathbf{J}\cdot\mathbf{S} = \frac{1}{2}(J^2 + S^2 - L^2)$:

$$\frac{\mathbf{J}\cdot\mathbf{S}}{J^2} = \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

Therefore:

$$\boxed{g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} = \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)}.}$$

E1 selection rules (LS): $\Delta L = 0, \pm 1$ (not $0 \to 0$); $\Delta S = 0$; $\Delta J = 0, \pm 1$ (not $0 \to 0$); $\Delta M_J = 0, \pm 1$ (not $0 \to 0$ if $\Delta J = 0$); parity changes.

Mercury Zeeman analysis ([8])

Zeeman shift: $\Delta E = g_J \mu_B B M_J$. Frequency shift per $\Delta M_J = 1$ at $B = 1\,\mathrm{T}$:

$$\Delta\nu_0 = \mu_B B/h = \frac{9.274 \times 10^{-24}}{6.626 \times 10^{-34}}\,\mathrm{Hz} = 14.0\,\mathrm{GHz}.$$

Component frequency shifts: $\Delta\nu = (g_u M_u - g_l M_l)\,\Delta\nu_0$ with $\Delta M = M_u - M_l = 0, \pm 1$.

Line A (3 components: $0, \pm 28$ GHz). Three components $\Rightarrow$ normal-Zeeman pattern $\Rightarrow$ either $S = 0 \to S = 0$ (singlet) with $g_u = g_l = 1$ producing shifts $\pm\Delta\nu_0 = \pm 14\,\mathrm{GHz}$ — but observed $\pm 28\,\mathrm{GHz}$ is $2\Delta\nu_0$. Therefore $g_u = g_l \equiv g$ with $g\,\Delta\nu_0 = 28\,\mathrm{GHz}$:

$$g_A = 28/14 = 2.$$

Three-component pattern with equal g's and $\Delta J \geq 1$ requires $J_u = J_l$ or $J_u = J_l \pm 1$ with degenerate σ shifts. Equal $g_u = g_l = 2$ identifies both as ${}^3S_1$ or related. Combined with the deduction below ($\lambda_A = 405$ nm is upper $\to J = 0$ component), we identify line A as $J_u = 1 \to J_l = 0$, which gives 3 components $0, \pm g_u \Delta\nu_0$. Hence $g_u = 2$, g_l undefined ($J_l = 0$).

Line B (6 components: $\pm 7, \pm 21, \pm 28$ GHz). Six components $\Rightarrow$ anomalous, $g_u \neq g_l$, and $J_u = J_l = 1$ would give 9; $J_u \to J_l$ with $J_u \to J_l \pm 1$ gives 6 lines if one g small. Pattern is $\pm(g_u - g_l)/2 \cdot 2\Delta\nu_0$ etc. Step size 7 GHz $= \Delta\nu_0/2$. Shifts: $g_u M_u - g_l M_l$ takes values $\pm\frac{1}{2}, \pm\frac{3}{2}, \pm 2$ in units of $\Delta\nu_0 = 14$ GHz. From the Zeeman pattern of $J_u = 2 \to J_l = 1$: shifts (in units $\Delta\nu_0$) for σ_+ ($\Delta M = +1$): $g_u(M_l+1) - g_l M_l = g_u + (g_u - g_l)M_l$ with $M_l = -1, 0, +1$: $g_u - (g_u - g_l), g_u, g_u + (g_u - g_l)$.

Solve: $g_u = 3/2$ (so $g_u \Delta\nu_0 = 21$ GHz), and $g_u - g_l = \pm 1/2$ giving σ outer shifts 14 GHz or 28 GHz and inner 21$\pm$ etc.

Match to observed $\pm 7, \pm 21, \pm 28$: try $g_u = 3/2$, $g_l = 1/2$: σ_+: $3/2 - 1 \cdot (-1) + \ldots$ — compute directly. Shifts $g_u M_u - g_l M_l$:

$\Delta M = +1$ (3 lines, $M_l = -1, 0, 1$): $g_u \cdot 0 - g_l(-1) = +1/2$; $g_u \cdot 1 - g_l \cdot 0 = +3/2$; $g_u \cdot 2 - g_l \cdot 1 = 3 - 1/2 = +5/2$.

$\Delta M = -1$ (3 lines): mirror $-1/2, -3/2, -5/2$.

$\Delta M = 0$ (3 lines, $M = -1, 0, +1$): $-(g_u - g_l), 0, +(g_u - g_l) = -1, 0, +1$.

But $\Delta M = 0$ is forbidden when $J_u = J_l$? Here $J_u = 2 \neq J_l = 1$ so allowed. Total 9 components, not 6. So line B is NOT $2 \to 1$ both with these g's.

Try $J_u \to J_l$ with $\Delta J = 1$ but $\Delta M = 0$ has zero shift only if $g_u = g_l$. Six components $\Rightarrow$ either $J_u = J_l$ (gives 3 $\sigma_\pm$ pairs = 6 lines if $\Delta M = 0$ central degenerate) or accidental degeneracy.

Cleanest reading: line B has $J_u = 1 \to J_l = 1$, $\Delta J = 0$ ($\Delta M = 0$ allowed except $0 \to 0$), $g_u \neq g_l$. Components: $\Delta M = +1$: $g_u(M_l+1) - g_l M_l$ with $M_l = -1, 0$: $g_u - g_l \cdot (-1) = g_u + g_l$ at $M_l = -1$? No: $g_u \cdot 0 - g_l(-1) = g_l$; $g_u \cdot 1 - g_l \cdot 0 = g_u$. Plus mirror gives $\pm g_l, \pm g_u$, total 4. Add $\Delta M = 0$ ($M = \pm 1$): $\pm(g_u - g_l)$, 2 lines. Total = 6 ✓.

Match $\pm 7, \pm 21, \pm 28$: $g_l \Delta\nu_0 = 7$ GHz, $g_u \Delta\nu_0 = 21$ GHz, $(g_u - g_l)\Delta\nu_0 = 14$ — but observed is 28 GHz, not 14. Try $g_l \Delta\nu_0 = 7$, $g_u \Delta\nu_0 = 21$, then $g_u + g_l$? Pairs $\pm 7, \pm 21, \pm(g_u - g_l)\Delta\nu_0 = \pm 14$. Doesn't match ± 28.

Alternative: $g_u = 2$, $g_l = 1/2$, $\Delta J = 0$, $J_u = J_l = 1$: σ: $g_l = 0.5 \to 7$, $g_u = 2 \to 28$; π: $(g_u - g_l) = 1.5 \to 21$. Components $\pm 7, \pm 28, \pm 21$ ✓.

$$\boxed{g_u(B) = 2, \quad g_l(B) = 1/2, \quad J_u = J_l = 1.}$$

Combined identification. All three lines share the same upper and lower *terms*. Line A has $J_u \to J_l = 0$ (3 components, $g_u = 2$), line B has $J_u = J_l = 1$ ($g_u = 2, g_l = 1/2$), line C has 9 components ($J_u = 2 \to J_l = 1$).

Same upper term with $g_u = 2$ and accommodating $J_u = 2, 1, 0, 1$ across A,B,C $\Rightarrow$ upper term ${}^3P_{2,1,0}$? But $g({}^3P_2) = 3/2$, $g({}^3P_1) = 3/2$, $g({}^3P_0)$ undefined. Doesn't give $g_u = 2$.

$g({}^3S_1) = 2$ ✓for upper. But ${}^3S_1$ has only $J = 1$. So $J_u = 1$ for all three lines. Then $J_l = 0, 1, 2$ across A,B,C respectively. Lower term must contain $J = 0, 1, 2 \Rightarrow {}^3P_{0,1,2}$ with $g({}^3P_0)$ undefined, $g({}^3P_1) = 3/2$, $g({}^3P_2) = 3/2$. But line B requires $g_l = 1/2$. Mismatch.

Reverse upper/lower: *lower* term ${}^3S_1$, upper term ${}^3P_{0,1,2}$. Then line A: $J_u = 0, J_l = 1$ (3 components, g_u irrelevant, $g_l = 2$ gives σ at $\pm g_l \Delta\nu_0 = \pm 28$ ✓).

Line B: $J_u = 1, J_l = 1$, $g_u = g({}^3P_1) = 3/2$, $g_l = 2$. Components: σ at $\pm g_u, \pm g_l = \pm 21, \pm 28$, π at $\pm(g_u - g_l) = \pm(-1/2) = \pm 7$. Total 6 lines $\pm 7, \pm 21, \pm 28$ ✓.

Line C: $J_u = 2, J_l = 1$, $g_u = g({}^3P_2) = 3/2$, $g_l = 2$. Components: $\Delta M = \pm 1$: $g_u M_u - g_l M_l$ with $M_u - M_l = \pm 1$, gives 5 outer lines; $\Delta M = 0$: 3 lines. Specifically shifts (in units $\Delta\nu_0 = 14$ GHz): $\Delta M = +1$, $M_l = -1, 0, +1, +2$ implies $M_u = 0, 1, 2, 3$ but $|M_u| \leq 2$, so $M_l = -1, 0, 1$: shifts $g_u \cdot 0 - g_l(-1) = 2$; $g_u - 0 = 3/2$; $g_u \cdot 2 - g_l = 3 - 2 = 1$. Mirror: $-1, -3/2, -2$.

$\Delta M = 0$, $M = \pm 1, \pm 2$ (omit $0 \to 0$ allowed since $J_u \neq J_l$): $(g_u - g_l)M = (-1/2)M$: $\mp 1/2, \mp 1$. Distinct values $\pm 1/2, \pm 1$.

Combined distinct shifts (units $\Delta\nu_0$): $\pm 1/2, \pm 1, \pm 3/2, \pm 2$. In GHz: $\pm 7, \pm 14, \pm 21, \pm 28$ ✓(with central component at 0 from $\Delta M = 0$, $M_u = M_l = 0$, but $0 \to 0$ forbidden since $J_u \neq J_l$? Allowed when $J_u \neq J_l$). 9 components total ✓.

$$\boxed{\text{Lower term: } {}^3S_1\ (L_l = 0, S_l = 1). \quad \text{Upper term: } {}^3P_{0,1,2}\ (L_u = 1, S_u = 1).}$$

$$J_A = 0 \to 1, \quad J_B = 1 \to 1, \quad J_C = 2 \to 1.$$

$$g_A^{(\text{lower})} = 2,\ g_A^{(\text{upper})} \text{ undefined}; \quad g_B^{(\text{upper})} = 3/2,\ g_B^{(\text{lower})} = 2.$$

Quality of LS-coupling in Hg ([4])

Hg ($Z = 80$) is heavy: spin-orbit coupling is comparable to electrostatic terms, so LS-coupling is at best approximate (jj-coupling is closer for inner shells). Evidence: (i) intercombination line $6^3P_1 \to 6^1S_0$ at 254 nm is strong despite formal $\Delta S = 0$ rule; (ii) singlet-triplet term separations not vastly larger than triplet fine-structure — indicating residual mixing; (iii) observed Landé g-factors $g_u = 2, g_l = 2$ etc. nevertheless agree with LS predictions ($g(^3S_1) = 2$, $g(^3P_1) = 3/2$) to good accuracy, so intermediate coupling preserves LS labelling for these levels.

Question 3 — Diatomic CO: rotation, vibration, Morse

Problem. Interpret the eigenenergy expansion, deduce I from the R-branch spacing, derive $\tilde{\nu}_v$ for the Morse potential, and determine β, r_0 for CO.

Physical origins ([5])

E_e: electronic energy in Born-Oppenheimer approximation (clamped nuclei).

$(n + \frac{1}{2})\hbar\omega_v$: vibrational energy from quantising small oscillations about the minimum of $V_e(r)$ at r_0, $\omega_v = \sqrt{V''(r_0)/\mu}$.

$hcBK(K+1)$: rigid-rotor rotational energy with rotational constant $B = \hbar/(4\pi c\mu r_0^2)$ from $E_K = \hbar^2 K(K+1)/(2I)$, $I = \mu r_0^2$.

Moment of inertia from R-branch spacing ([5])

Transition $\Delta n = +1$, $\Delta K = -1$ (P-branch). Energy of line from upper $(n+1, K-1)$ to lower (n, K):

$$\tilde{\nu} = \tilde{\nu}_v + B[(K-1)K - K(K+1)] = \tilde{\nu}_v - 2BK, \qquad K = 1, 2, 3, \ldots$$

Successive spacing: $|\Delta\tilde{\nu}| = 2B$.

Observed first three: $\tilde{\nu}_v - 3.9, \tilde{\nu}_v - 7.8, \tilde{\nu}_v - 11.7$. Spacing $3.9\,\text{cm}^{-1}$. Hence:

$$2B = 3.9\,\text{cm}^{-1}, \qquad B = 1.95\,\text{cm}^{-1}.$$

Convert: B in SI $= Bc = 1.95 \times 10^2\,\text{m}^{-1} \times 3 \times 10^8\,\text{m}\,\text{s}^{-1} = 5.85 \times 10^{10}\,\text{Hz}$. Actually hcB has units of energy; we use $I = \hbar/(4\pi cB)$ with B in m^{-1}:

$$I = \frac{\hbar}{4\pi cB} = \frac{1.055 \times 10^{-34}}{4\pi \times 3 \times 10^8 \times 195}\,\text{kg}\,\text{m}^2.$$

Evaluate denominator: $4\pi \times 3 \times 10^8 \times 195 = 7.35 \times 10^{11}$:

$$I = \frac{1.055 \times 10^{-34}}{7.35 \times 10^{11}}\,\text{kg}\,\text{m}^2 = \boxed{1.44 \times 10^{-46}\,\text{kg}\,\text{m}^2}.$$

Why diatomic molecules bind; Morse derivation ([9])

Binding: At intermediate r the attractive part dominates — valence electrons delocalise over both nuclei (covalent overlap) lowering kinetic energy and forming bonding molecular orbitals. At small r, Pauli exclusion of core electrons and nuclear Coulomb repulsion produce a steep wall. The competition yields a minimum at r_0. The Morse potential models the soft attractive tail (asymptote $V \to D$, dissociation) and the steep repulsive core, but with finite slope at small r.

Morse parameters in $\tilde{\nu}_v$. Expand about r_0:

$$V(r) = D\big[1 - e^{-\beta(r-r_0)}\big]^2.$$

Derivative:

$$V'(r) = 2D\big[1 - e^{-\beta(r-r_0)}\big] \cdot \beta e^{-\beta(r-r_0)}.$$

At $r = r_0$: $V'(r_0) = 0$ ✓. Second derivative at r_0:

$$V''(r) = 2D\beta\big[\beta e^{-2\beta(r-r_0)} - \beta e^{-\beta(r-r_0)}\big(1 - e^{-\beta(r-r_0)}\big)\big] \cdot 2.$$

Cleaner: differentiate V' once more directly. Let $u = e^{-\beta(r-r_0)}$:

$$V = D(1-u)^2, \qquad V' = 2D(1-u)(-u')(-1)/\beta \cdot \beta \ldots$$

Use $V = D(1-u)^2$ with $u(r_0) = 1$. $V' = -2D(1-u)u' = -2D(1-u)\cdot(-\beta u) = 2D\beta u(1-u)$. Differentiate:

$$V'' = 2D\beta[u'(1-u) + u(-u')] = 2D\beta \cdot (-\beta u)(1-2u).$$

At r_0: $u = 1$, $1 - 2u = -1$:

$$V''(r_0) = 2D\beta \cdot (-\beta)(-1) = 2D\beta^2.$$

Harmonic frequency:

$$\omega_v = \sqrt{V''(r_0)/\mu} = \beta\sqrt{2D/\mu}.$$

Wavenumber:

$$\boxed{\tilde{\nu}_v = \frac{\omega_v}{2\pi c} = \frac{\beta}{2\pi c}\sqrt{\frac{2D}{\mu}}.}$$

Determine β, r_0 ([6])

$E_{\text{diss}} = D$ (Morse asymptote $V(\infty) = D$, $V(r_0) = 0$). Thus $D = 11.16\,\text{eV} = 1.788 \times 10^{-18}\,\text{J}$.

Reduced mass: $\mu = (12 \times 16)/(12+16)\,\text{u} = 6.857\,\text{u} = 6.857 \times 1.661 \times 10^{-27}\,\text{kg}$:

$$\mu = 1.139 \times 10^{-26}\,\text{kg}.$$

β from $\tilde{\nu}_v$: $\tilde{\nu}_v = 2143\,\text{cm}^{-1} = 2.143 \times 10^5\,\text{m}^{-1}$. Invert:

$$\beta = 2\pi c\tilde{\nu}_v\sqrt{\mu/(2D)}.$$

Compute $2\pi c\tilde{\nu}_v = \omega_v$:

$$\omega_v = 2\pi \times 3 \times 10^8 \times 2.143 \times 10^5\,\text{rad}\,\text{s}^{-1} = 4.040 \times 10^{14}\,\text{rad}\,\text{s}^{-1}.$$

Compute $\sqrt{\mu/(2D)}$:

$$\mu/(2D) = 1.139 \times 10^{-26}/(3.576 \times 10^{-18}) = 3.185 \times 10^{-9}\,\text{s}^2\,\text{m}^{-2}.$$

$$\sqrt{\mu/(2D)} = 5.644 \times 10^{-5}\,\mathrm{s\,m^{-1}}.$$

Thus:

$$\beta = 4.040 \times 10^{14} \times 5.644 \times 10^{-5}\,\mathrm{m^{-1}} = \boxed{2.28 \times 10^{10}\,\mathrm{m^{-1}}}.$$

r_0 **from** $I = \mu r_0^2$**:**

$$r_0 = \sqrt{I/\mu} = \sqrt{1.44 \times 10^{-46}/1.139 \times 10^{-26}}\,\mathrm{m}.$$

$$r_0 = \sqrt{1.265 \times 10^{-20}}\,\mathrm{m} = \boxed{1.125 \times 10^{-10}\,\mathrm{m}} = 0.113\,\mathrm{nm}.$$

Product:

$$\beta r_0 = 2.28 \times 10^{10} \times 1.125 \times 10^{-10} = \boxed{2.57}.$$

Comment. $\beta r_0 \sim O(1)$ confirms the natural Morse length $\beta^{-1} \sim r_0$: the potential well width is comparable to the bond length, consistent with strong binding ($D \gg \hbar\omega_v$: harmonic region small fraction of well depth, $\hbar\omega_v/D = 0.024$).

Question 4 — Lasers: gain, threshold, broadening

Problem. Compare 3- and 4-level thresholds, derive gain coefficient, obtain $\sigma_{21}^{\max}$, and apply to ruby and Ti:sapphire.

3-level vs 4-level threshold ([4])

3-level (e.g. ruby): lower laser level is the ground state, population $N_1 \sim N_{\text{ions}}$. Inversion $N_2 > N_1$ requires pumping more than half the ions into upper level: $N_2 > N_{\text{ions}}/2$. Pump energy threshold $\propto N_{\text{ions}} V$.

4-level: lower laser level rapidly drained (τ_1 short), $N_1 \approx 0$ in steady state. Inversion attained for any $N_2 > 0$: N_2 only needs to compensate cavity losses. Pump threshold $\propto$ small loss-determined inversion only.

Hence 3-level threshold is $\sim 10^4$–$10^6\times$ larger.

Gain coefficient ([7])

Stimulated emission rate per atom: $W_{21} = B_{21}\rho(\omega)g(\omega - \omega_0)$.

Photon flux $I/(\hbar\omega)$ relates to ρ: for a quasi-monochromatic plane wave $\rho(\omega)\,\mathrm{d}\omega = (I/c)\,\mathrm{d}\omega$ on resonance. Use directional radiation $\rho(\omega) = Ig(\omega - \omega_0)/c$ (radiation field with line profile narrower than g, so per unit angular-frequency interval $\rho = I/c$ at ω).

Net rate of photon production per unit volume:

$$\frac{\mathrm{d}I}{\mathrm{d}z} \cdot \frac{1}{\hbar\omega} = (N_2 B_{21} - N_1 B_{12})\,\rho(\omega)\,g(\omega - \omega_0).$$

Use $g_1 B_{12} = g_2 B_{21}$:

$$\frac{\mathrm{d}I}{\mathrm{d}z} = \hbar\omega\, B_{21}\left(N_2 - \frac{g_2}{g_1}N_1\right) g(\omega - \omega_0)\,\rho(\omega).$$

Substitute $\rho = Ig/c$? More carefully, gain coefficient γ defined by $\mathrm{d}I/\mathrm{d}z = \gamma I$. Combine with $A_{21} = \hbar\omega^3 B_{21}/(\pi^2 c^3)$:

$$B_{21} = \frac{\pi^2 c^3}{\hbar\omega^3} A_{21}.$$

Standard result for gain coefficient:

$$\boxed{\gamma(\omega) = \frac{\pi^2 c^2}{\omega^2}\, A_{21}\, g(\omega-\omega_0)\left(N_2 - \frac{g_2}{g_1}N_1\right).}$$

Validity: (i) two-level rate-equation regime (no coherence, $\Delta t \gg$ dephasing time); (ii) homogeneously narrow probe $\Delta\omega_{\text{probe}} \ll$ linewidth; (iii) weak field (no saturation: stimulated rate $\ll$ relaxation rate); (iv) $g(\omega-\omega_0)$ describes the homogeneous medium response.

Natural broadening, $\sigma_{21}^{\max}$ ([5])

Natural broadening: exponential decay → Lorentzian profile. FWHM:

$$\Delta\omega_{\text{nat}} = \frac{1}{\tau_2} + \frac{1}{\tau_1}.$$

Lorentzian peak value: $g(0) = 2/(\pi\,\Delta\omega_{\text{nat}})$.
Cross-section $\sigma_{21}(\omega) = \gamma/(N_2 - (g_2/g_1)N_1) = (\pi^2c^2/\omega^2)A_{21}g(\omega-\omega_0)$:

$$\sigma_{21}(\omega_0) = \frac{\pi^2 c^2}{\omega_0^2}\, A_{21}\, \frac{2}{\pi\Delta\omega_{\text{nat}}}.$$

Maximum: $A_{21} \le \Delta\omega_{\text{nat}}$ (since $1/\tau_2$ contributes to both A_{21} and width); maximum when $A_{21} = \Delta\omega_{\text{nat}}$ (radiative limit, $\tau_1 \to \infty$):

$$\sigma_{21}^{\max} = \frac{\pi^2 c^2}{\omega_0^2}\cdot\frac{2}{\pi} = \boxed{\frac{2\pi c^2}{\omega_0^2}} = \frac{\lambda^2}{2\pi}.$$

At $\lambda = 700\,\text{nm}$:

$$\sigma_{21}^{\max} = \frac{(7\times10^{-7})^2}{2\pi}\,\text{m}^2 = \frac{4.9\times10^{-13}}{6.283}\,\text{m}^2 = \boxed{7.8\times10^{-14}\,\text{m}^2}.$$

(a) Ruby 3-level laser

Threshold: $\gamma L \ge \frac{1}{2}|\ln(R_1R_2)|$. With $R_1 = 1, R_2 = 0.95$, $L = 4\,\text{cm}$:

$$\gamma_{\text{th}} = \frac{|\ln 0.95|}{2L} = \frac{0.0513}{0.08}\,\text{m}^{-1} = 0.641\,\text{m}^{-1}.$$

3-level inversion: $N_2 - N_1 = 2N_2 - N_{\text{ions}}$. At threshold $N_2 \approx N_{\text{ions}}/2 + \Delta N_{\text{th}}/2$. Pump must promote $N_{\text{ions}}/2$ ions (the small extra ΔN_{th} is negligible):

$$N_{\text{pump}} \approx N_{\text{ions}}/2 = 2\times10^{19}\,\text{cm}^{-3} = 2\times10^{25}\,\text{m}^{-3}.$$

Volume: $V = \pi(0.2\,\text{cm})^2 \times 4\,\text{cm} = 0.503\,\text{cm}^3 = 5.03\times10^{-7}\,\text{m}^3$.
Number of pump photons absorbed: $N_{\text{ph}} = N_{\text{pump}}\cdot V$:

$$N_{\text{ph}} = 2\times10^{25}\times5.03\times10^{-7} = 1.01\times10^{19}.$$

Pump photon energy at 500 nm: $E_{\text{ph}} = hc/\lambda$:

$$E_{\text{ph}} = \frac{6.626\times10^{-34}\times3\times10^8}{5\times10^{-7}}\,\text{J} = 3.98\times10^{-19}\,\text{J}.$$

Total pump energy:

$$E_{\text{pump}} = N_{\text{ph}}\cdot E_{\text{ph}} = 1.01\times10^{19}\times3.98\times10^{-19}\,\text{J}.$$

$$\boxed{E_{\text{pump}}^{(\text{ruby})} \approx 4.0\,\text{J}.}$$

(b) Ti:sapphire 4-level laser

Threshold inversion. Cavity condition $\gamma_{\text{th}} = |\ln(R_1 R_2)|/(2L) = 0.0513/0.02 = 2.56\,\text{m}^{-1}$.
With $g_2 = g_1$: $N_2 - N_1 = N_2$ (4-level, $N_1 \approx 0$). $\gamma = \sigma_{21}\Delta N$:

$$\Delta N_{\text{th}} = \gamma_{\text{th}}/\sigma_{21} = 2.56/(3 \times 10^{-23})\,\text{m}^{-3}.$$

$$\sigma_{21} = 3 \times 10^{-19}\,\text{cm}^2 = 3 \times 10^{-23}\,\text{m}^2.$$

$$\Delta N_{\text{th}} = 8.55 \times 10^{22}\,\text{m}^{-3} = 8.55 \times 10^{16}\,\text{cm}^{-3}.$$

Pump volume. Pumped along rod: $V_{\text{pump}} = \pi(50\,\mu\text{m})^2 \times 1\,\text{cm}$:

$$V_{\text{pump}} = \pi \times 2.5 \times 10^{-9}\,\text{m}^2 \times 10^{-2}\,\text{m} = 7.85 \times 10^{-11}\,\text{m}^3.$$

Photons needed: $N_{\text{ph}} = \Delta N_{\text{th}} \cdot V_{\text{pump}}$:

$$N_{\text{ph}} = 8.55 \times 10^{22} \times 7.85 \times 10^{-11} = 6.71 \times 10^{12}.$$

Pump-photon energy at 532 nm: $E_{\text{ph}} = hc/\lambda = 3.74 \times 10^{-19}$ J:

$$E_{\text{pulse}} = 6.71 \times 10^{12} \times 3.74 \times 10^{-19}\,\text{J}.$$

$$\boxed{E_{\text{pulse}}^{(\text{Ti:S})} \approx 2.5\,\mu\text{J}.}$$

cw threshold power. Each pump photon excites one ion which decays at rate $1/\tau_2$. To maintain ΔN_{th}, pump rate per unit volume $= \Delta N_{\text{th}}/\tau_2$:

$$P_{\text{cw}} = \Delta N_{\text{th}} \cdot V_{\text{pump}} \cdot E_{\text{ph}}/\tau_2 = E_{\text{pulse}}/\tau_2.$$

Substitute $\tau_2 = 3 \times 10^{-6}$ s:

$$P_{\text{cw}} = \frac{2.51 \times 10^{-6}}{3 \times 10^{-6}}\,\text{W} = \boxed{0.84\,\text{W}}.$$

Natural linewidth.

$$\Delta\omega_{\text{nat}} = 1/\tau_2 + 1/\tau_1 = 1/3 \times 10^{-6} + 1/10^{-10} = 3.3 \times 10^5 + 10^{10}\,\text{s}^{-1}.$$

Dominant term $1/\tau_1$:

$$\Delta\nu_{\text{nat}} = \frac{\Delta\omega_{\text{nat}}}{2\pi} = \frac{10^{10}}{2\pi}\,\text{Hz} = \boxed{1.6\,\text{GHz}}.$$

Other broadening. Ti:sapphire emission band spans 300 nm ($\sim$ 180 THz) — vastly larger than 1.6 GHz. Vibronic phonon coupling in the sapphire host produces strong inhomogeneous broadening (configuration coordinate model, host-lattice phonon sidebands). Doppler is negligible (solid medium). Hence inhomogeneous (vibronic / phonon-induced) broadening dominates by $\sim 10^5$.